

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2022 P2H Worked Solutions

Jun 2022 | P2H

29 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3**TOPIC: **Statistics Toolkit**

Jun 2022 P2H - Jun 2022 | P2H

- 1 Here are some integers where $a < b < c < d$

 $a \quad b \quad c \quad d \quad d \quad d$

The mode of the integers is 9

The median of the integers is 8

The range of the integers is 4

Work out the value of a , the value of b , the value of c and the value of d

 $a = \dots\dots\dots$ $b = \dots\dots\dots$ $c = \dots\dots\dots$ $d = \dots\dots\dots$

(Total for Question 1 is 3 marks)

Dr. Eslam

PAST PAPER SOLUTION**Q#: 1****Marks: 3**TOPIC: **Statistics Toolkit**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Statistics Toolkit. The tag is correct.**Solution**

The mode is 9, so

$$d = 9$$

The range is 4:

$$9 - a = 4$$

$$a = 5$$

The median is 8, so

$$\frac{c + d}{2} = 8$$

$$\frac{c + 9}{2} = 8$$

$$c = 7$$

The integers are in order and $a < b < c$, so

$$b = 6$$

FINAL ANSWER $a = 5, b = 6, c = 7, d = 9.$

PAST PAPER QUESTION**Q#: 2****Marks: 4****TOPIC: Graphing Inequalities**

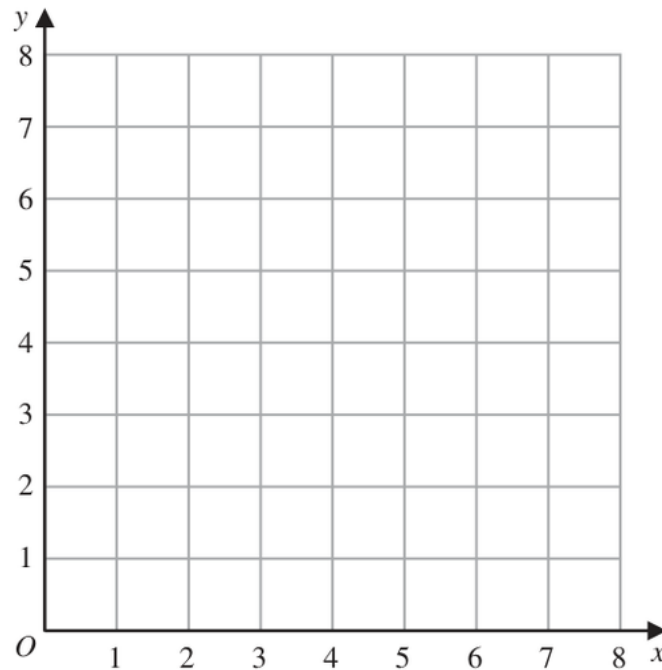
Jun 2022 P2H - Jun 2022 | P2H

2 (a) On the grid, draw and label with its equation the straight line with equation

(i) $y = 1$

(ii) $x = 2$

(iii) $x + y = 7$



(3)

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$y \geq 1$

$x \geq 2$

$x + y \leq 7$

Label the region **R**.

(1)

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION**Q#: 2****Marks: 4****TOPIC: Graphing Inequalities**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct. This is drawing and shading inequalities.**Solution**

Draw the boundary lines:

$$y = 1, \quad x = 2, \quad x + y = 7$$

The required region is

$$y \geq 1$$

$$x \geq 2$$

$$x + y \leq 7$$

FINAL ANSWERShade $y \geq 1$, $x \geq 2$, $x + y \leq 7$.

PAST PAPER QUESTION**Q#: 3****Marks: 3****TOPIC: Standard & Compound Units**

Jun 2022 P2H - Jun 2022 | P2H

- 3** An aeroplane travelled from New York City to Los Angeles.

The aeroplane travelled a distance of 3980 kilometres in 5 hours 24 minutes.

Work out the average speed of the aeroplane.

Give your answer in kilometres per hour correct to the nearest whole number.

..... kilometres per hour

(Total for Question 3 is 3 marks)

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PAST PAPER SOLUTION**Q#: 3****Marks: 3****TOPIC: Standard & Compound Units**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to Standard & Compound Units.**Solution**

The time is

$$5 \text{ h } 24 \text{ min} = 5.4 \text{ h}$$

Average speed:

$$3980 \div 5.4 = 737.037 \dots$$

FINAL ANSWER

737 kilometres per hour.

PAST PAPER QUESTION**Q#: 4****Marks: 3**TOPIC: **Fractions**

Jun 2022 P2H - Jun 2022 | P2H

4 Show that $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$

(Total for Question 4 is 3 marks)

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PAST PAPER SOLUTION**Q#: 4****Marks: 3**TOPIC: **Fractions**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Fractions. The tag is correct.**Method**

$$5\frac{1}{3} = \frac{16}{3}$$

$$2\frac{6}{7} = \frac{20}{7}$$

Use a common denominator:

$$\begin{aligned}\frac{16}{3} - \frac{20}{7} &= \frac{112}{21} - \frac{60}{21} \\ &= \frac{52}{21} = 2\frac{10}{21}\end{aligned}$$

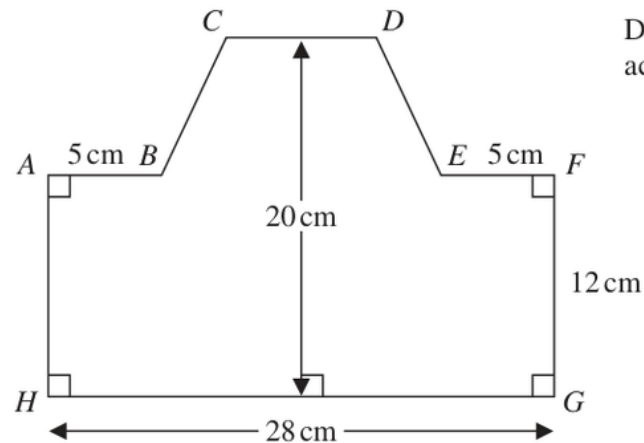
FINAL ANSWER

$$2\frac{10}{21}$$

PAST PAPER QUESTION**Q#: 5****Marks: 4**TOPIC: **Area & Perimeter**

Jun 2022 P2H - Jun 2022 | P2H

- 5 The diagram shows an 8-sided shape $ABCDEFGH$.

Diagram **NOT**
accurately drawn

$$HG = 28 \text{ cm} \quad FG = 12 \text{ cm} \quad AB = EF = 5 \text{ cm}$$

The height of the shape is 20 cm

 CD is parallel to HG The area of shape $ABCDEFGH$ is 434 cm^2 Find the length of CD .

..... cm

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 4**TOPIC: **Area & Perimeter**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Area & Perimeter. The tag is correct.**Solution**

The lower rectangle has width 28 cm and height 12 cm:

$$28 \times 12 = 336 \text{ cm}^2$$

The top part is a trapezium of height

$$20 - 12 = 8 \text{ cm}$$

The lower parallel side of this trapezium is

$$28 - 5 - 5 = 18 \text{ cm}$$

Area of the trapezium:

$$434 - 336 = 98 \text{ cm}^2$$

Use

$$\text{area} = \frac{1}{2}(a + b)h$$

$$98 = \frac{1}{2}(18 + CD)(8)$$

$$98 = 4(18 + CD)$$

$$18 + CD = 24.5$$

$$CD = 6.5$$

FINAL ANSWER

6.5 cm.

PAST PAPER QUESTION**Q#: 6****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2022 P2H - Jun 2022 | P2H

- 6 The diagram shows triangle PQR .

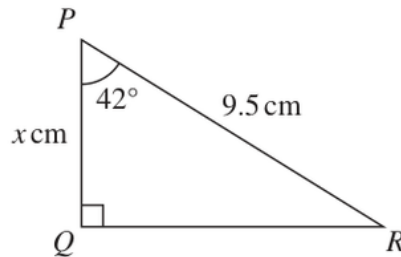


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION**Q#: 6** **Marks: 3****TOPIC:** Right-Angled Triangles - Pythagoras & Trigonometry

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to Right-Angled Triangles - Pythagoras & Trigonometry. This is a right-angled trigonometry problem.

Solution

In the right-angled triangle, x is adjacent to the 42° angle and 9.5 cm is the hypotenuse.

$$\cos 42^\circ = \frac{x}{9.5}$$

$$x = 9.5 \cos 42^\circ = 7.059 \dots$$

FINAL ANSWER

7.1 cm.

PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Standard & Compound Units**

Jun 2022 P2H - Jun 2022 | P2H

7 Change a speed of 81 kilometres per hour to a speed in metres per second.

..... metres per second

(Total for Question 7 is 3 marks)

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PAST PAPER SOLUTION**Q#: 7****Marks: 3**TOPIC: **Standard & Compound Units**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct. This is converting kilometres per hour to metres per second.**Solution**

To convert km/h to m/s, divide by 3.6.

$$81 \div 3.6 = 22.5$$

FINAL ANSWER

22.5 m/s.

PAST PAPER QUESTION**Q#: 8****Marks: 5**TOPIC: **Percentages**

Jun 2022 P2H - Jun 2022 | P2H

- 8 Behnaz makes 300 celebration cards so that

$$\begin{array}{l} \text{number of} \\ \text{birthday cards} \end{array} : \begin{array}{l} \text{number of} \\ \text{anniversary cards} \end{array} : \begin{array}{l} \text{number of} \\ \text{congratulations cards} \end{array} = 7:5:3$$

$\frac{2}{5}$ of the birthday cards have numbers on them.

36% of the anniversary cards have numbers on them.

None of the congratulations cards have numbers on them.

Work out what fraction of the 300 cards have numbers on them.

Give your answer in its simplest form.

(Total for Question 8 is 5 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 5****TOPIC: Percentages**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Percentages. The tag is correct.**Method**

The ratio 7 : 5 : 3 has 15 parts.

Birthday cards:

$$\frac{7}{15} \times 300 = 140$$

Anniversary cards:

$$\frac{5}{15} \times 300 = 100$$

Congratulations cards:

$$\frac{3}{15} \times 300 = 60$$

Birthday cards with numbers:

$$\frac{2}{5} \times 140 = 56$$

Anniversary cards with numbers:

$$36\% \times 100 = 36$$

Congratulations cards with numbers:

$$0$$

Total with numbers:

$$56 + 36 = 92$$

Fraction of all cards:

$$\frac{92}{300} = \frac{23}{75}$$

FINAL ANSWER

$$\frac{23}{75}$$

PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jun 2022 P2H - Jun 2022 | P2H

- 9 Pasha invests 50 000 dollars in a savings account for 4 years.
He gets 1.3% per year compound interest.

Work out how much money Pasha will have in his savings account at the end of 4 years.
Give your answer correct to the nearest dollar.

..... dollars

(Total for Question 9 is 3 marks)

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PAST PAPER SOLUTION**Q#: 9****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Compound Interest and Depreciation. The tag is correct.**Method**

The multiplier for 1.3% compound interest is

$$1.013$$

After 4 years,

$$50000(1.013)^4 = 52651.1408\dots$$

Correct to the nearest dollar,

$$52651.1408\dots \approx 52651$$

FINAL ANSWER

52651 dollars

PAST PAPER QUESTION**Q#: 10****Marks: 3****TOPIC: Simultaneous Equations**

Jun 2022 P2H - Jun 2022 | P2H

10 Solve the simultaneous equations

$$7x + 3y = 3$$

$$3x - y = 7$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 10 is 3 marks)

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PAST PAPER SOLUTION**Q#: 10****Marks: 3****TOPIC: Simultaneous Equations**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Simultaneous equations. The tag is correct.**Solution**

$$7x + 3y = 3$$

$$3x - y = 7$$

From the second equation:

$$y = 3x - 7$$

Substitute into the first equation:

$$7x + 3(3x - 7) = 3$$

$$7x + 9x - 21 = 3$$

$$16x = 24$$

$$x = \frac{3}{2}$$

Then

$$y = 3\left(\frac{3}{2}\right) - 7 = \frac{9}{2} - 7 = -\frac{5}{2}$$

FINAL ANSWER

$$x = \frac{3}{2}, y = -\frac{5}{2}.$$

PAST PAPER QUESTION**Q#: 11****Marks: 3**TOPIC: **Factorising**

Jun 2022 P2H - Jun 2022 | P2H

11 (i) Factorise $x^2 + 5x - 24$
(2)(ii) Hence, solve $x^2 + 5x - 24 = 0$
(1)

(Total for Question 11 is 3 marks)

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PAST PAPER SOLUTION**Q#: 11****Marks: 3****TOPIC: Factorising**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Factorising. The tag is correct.**Solution**

$$x^2 + 5x - 24 = (x + 8)(x - 3)$$

So

$$(x + 8)(x - 3) = 0$$

FINAL ANSWER

$$x = -8 \text{ or } x = 3$$

PAST PAPER QUESTION**Q#: 12****Marks: 3**TOPIC: **Statistics Toolkit**

Jun 2022 P2H - Jun 2022 | P2H

12 Larry is a delivery man.

He has 7 parcels to deliver.

The mean weight of the 7 parcels is 2.7 kg

Larry delivers 3 of the parcels.

Each of these 3 parcels has a weight of W kg

The mean weight of the other 4 parcels is 3.3 kg

Work out the value of W $W = \dots\dots\dots$

(Total for Question 12 is 3 marks)

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PAST PAPER SOLUTION**Q#: 12****Marks: 3**TOPIC: **Statistics Toolkit**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Statistics Toolkit. The tag is correct.**Solution**

Total mass of the 7 parcels:

$$7 \times 2.7 = 18.9$$

Total mass of the 4 remaining parcels:

$$4 \times 3.3 = 13.2$$

Total mass of the 3 delivered parcels:

$$18.9 - 13.2 = 5.7$$

Each delivered parcel has mass W :

$$3W = 5.7$$

$$W = 1.9$$

FINAL ANSWER

1.9 kg.

PAST PAPER QUESTION

Q#: 13 Marks: 7

TOPIC: Cumulative Frequency Diagrams

Jun 2022 P2H - Jun 2022 | P2H

13 The table gives information about the ages, in years, of 80 people in a train carriage.

Age (a years)	Frequency
$0 < a \leq 20$	7
$20 < a \leq 30$	25
$30 < a \leq 40$	20
$40 < a \leq 50$	14
$50 < a \leq 60$	8
$60 < a \leq 70$	6

(a) Complete the cumulative frequency table.

Age (a years)	Cumulative frequency
$0 < a \leq 20$	
$0 < a \leq 30$	
$0 < a \leq 40$	
$0 < a \leq 50$	
$0 < a \leq 60$	
$0 < a \leq 70$	

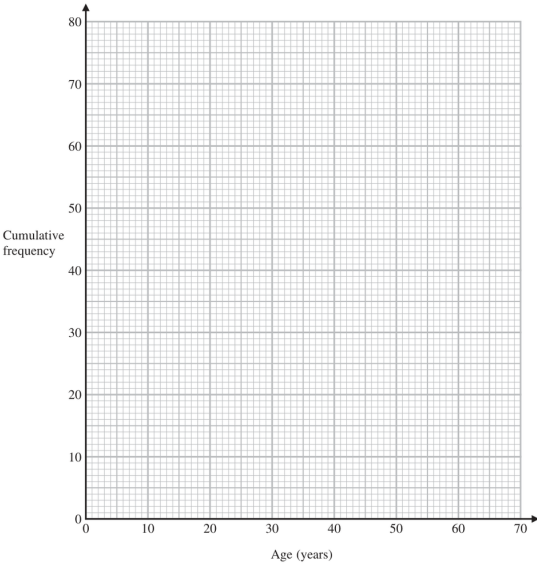
(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.

(2)

(c) Use your graph to find an estimate for the median age of the 80 people.

..... years
(1)



Of the people in the train carriage, 60% of those who are aged between 18 and 65 are going to work. None of the other people in the train carriage are going to work.

(d) Use your graph to find an estimate for the number of people in the train carriage who are going to work.

(3)

(Total for Question 13 is 7 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 7****TOPIC: Cumulative Frequency Diagrams**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Cumulative Frequency Diagrams. The tag is correct.**Solution**

Complete the cumulative frequencies:

7, 32, 52, 66, 74, 80

Plot the points

 $(0, 0), (20, 7), (30, 32), (40, 52), (50, 66), (60, 74), (70, 80)$

The median is at cumulative frequency 40.

This is between (30, 32) and (40, 52):

$$30 + \frac{40 - 32}{52 - 32} \times 10 = 34$$

So the median age is about 34 years.

Estimate the number aged between 18 and 65:

$$F(18) \approx \frac{18}{20} \times 7 = 6.3$$

$$F(65) \approx 74 + \frac{5}{10}(80 - 74) = 77$$

$$77 - 6.3 = 70.7$$

60% are going to work:

$$0.6(70.7) = 42.42$$

FINAL ANSWER

cumulative frequencies 7, 32, 52, 66, 74, 80, median about 34 years, about 42 people going to work.

PAST PAPER QUESTION**Q#: 14****Marks: 7****TOPIC: Rearranging Formulas**

Jun 2022 P2H - Jun 2022 | P2H

- 14** (a) Expand and simplify $(5 - x)(2x + 3)(x + 4)$
Show your working clearly.

.....
(3)

- (b) Make c the subject of $g = \frac{c + 3}{4 + c} - 7$

.....
(4)

(Total for Question 14 is 7 marks)

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PAST PAPER SOLUTION**Q#: 14****Marks: 7****TOPIC: Rearranging Formulas**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Rearranging formulas. The tag is correct.**Solution**

(a)

$$(5 - x)(2x + 3)(x + 4)$$

$$(2x + 3)(x + 4) = 2x^2 + 11x + 12$$

$$(5 - x)(2x^2 + 11x + 12) = -2x^3 - x^2 + 43x + 60$$

(b)

$$g = \frac{c + 3}{4 + c} - 7$$

$$g + 7 = \frac{c + 3}{c + 4}$$

$$(g + 7)(c + 4) = c + 3$$

$$c(g + 6) = -4g - 25$$

FINAL ANSWER(a) $-2x^3 - x^2 + 43x + 60$, (b) $c = \frac{-4g-25}{g+6}$

PAST PAPER QUESTION**Q#: 15****Marks: 7****TOPIC: Solving Linear Equations**

Jun 2022 P2H - Jun 2022 | P2H

15 (a) Solve $\frac{4x+5}{3} - \frac{3-2x}{2} = 13$

Show clear algebraic working.

$$x = \dots\dots\dots$$

(4)

(b) Solve the inequality $2y^2 - 7y - 30 \leq 0$
Show your working clearly.

$$\dots\dots\dots$$

(3)

(Total for Question 15 is 7 marks)

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PAST PAPER SOLUTION**Q#: 15****Marks: 7****TOPIC: Solving Linear Equations**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to solving linear equations. The four-mark linear equation is the largest part of the question.

Solution

(a)

$$\frac{4x + 5}{3} - \frac{3 - 2x}{2} = 13$$

Multiply by 6:

$$2(4x + 5) - 3(3 - 2x) = 78$$

$$8x + 10 - 9 + 6x = 78$$

$$14x + 1 = 78$$

$$14x = 77$$

$$x = \frac{77}{14} = \frac{11}{2}$$

(b)

$$2y^2 - 7y - 30 \leq 0$$

Factorise:

$$(2y + 5)(y - 6) \leq 0$$

The roots are $-\frac{5}{2}$ and 6. The quadratic is ≤ 0 between the roots.

FINAL ANSWER

$$x = \frac{11}{2}, \text{ and } -\frac{5}{2} \leq y \leq 6.$$

PAST PAPER QUESTION**Q#: 16****Marks: 8****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jun 2022 P2H - Jun 2022 | P2H

16 100 farmers are asked if they have goats (G), sheep (S) or chickens (C) on their farms.

Of these farmers

31 have sheep

53 have chickens

6 have goats, sheep and chickens

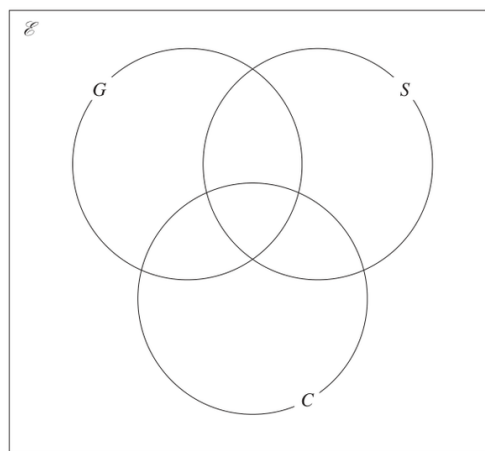
11 have sheep and goats

17 have sheep and chickens

18 have goats and chickens

20 do not have any goats, sheep or chickens

(a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.



(3)

(b) Find

(i) $n(G)$

(1)

(ii) $n([G \cup S]')$

(1)

(iii) $n(G' \cap C)$

(1)

One of the farmers who has chickens is chosen at random.

(c) Find the probability that this farmer also has goats.

(2)

(Total for Question 16 is 8 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 8****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Probability Diagrams - Venn & Tree Diagrams. The tag is correct.**Solution**

There are 100 farmers and 20 have none of the three animals, so 80 are in the Venn diagram.
 Pair-only regions:

$$S \cap G \text{ only} = 11 - 6 = 5$$

$$S \cap C \text{ only} = 17 - 6 = 11$$

$$G \cap C \text{ only} = 18 - 6 = 12$$

$$S \text{ only} = 31 - 5 - 11 - 6 = 9$$

$$C \text{ only} = 53 - 11 - 12 - 6 = 24$$

Then

$$G \text{ only} = 80 - (9 + 24 + 5 + 11 + 12 + 6) = 13$$

$$n(G) = 13 + 5 + 12 + 6 = 36$$

$$n((G \cup S)') = 20 + 24 = 44$$

$$n(G' \cap C) = 24 + 11 = 35$$

$$P(G | C) = \frac{18}{53}$$

FINAL ANSWER

36, 44, 35, $\frac{18}{53}$.

PAST PAPER QUESTION**Q#: 17****Marks: 4****TOPIC: Direct & Inverse Proportion**

Jun 2022 P2H - Jun 2022 | P2H

17 M varies directly as the cube of h

$$M = 4 \text{ when } h = 0.5$$

Find the value of h when $M = 500$

(Total for Question 17 is 4 marks)

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PAST PAPER SOLUTION**Q#: 17****Marks: 4****TOPIC: Direct & Inverse Proportion**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Direct and Inverse Proportion. The tag is correct.**Method** M varies directly as h^3 , so

$$M = kh^3$$

Use $M = 4$ when $h = 0.5$:

$$4 = k(0.5)^3$$

$$k = \frac{4}{0.125} = 32$$

So

$$M = 32h^3$$

When $M = 500$,

$$500 = 32h^3$$

$$h^3 = 15.625$$

$$h = 2.5$$

FINAL ANSWER $h = 2.5$.

PAST PAPER QUESTION**Q#: 18****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Jun 2022 P2H - Jun 2022 | P2H

18 $X = \frac{2a - b}{f}$

 $a = 7.5$ correct to 1 decimal place. $b = 3.42$ correct to 2 decimal places. $f = 2$ correct to the nearest whole number.

Work out the upper bound of the value of X
Show your working clearly.

(Total for Question 18 is 3 marks)

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PAST PAPER SOLUTION**Q#: 18****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Rounding, Estimation and Bounds. The tag is correct.**Method**

$$X = \frac{2a - b}{f}$$

For the upper bound, make the numerator large and the denominator small.

$$a < 7.55, \quad b \geq 3.415, \quad f \geq 1.5$$

$$X_{\text{upper}} = \frac{2(7.55) - 3.415}{1.5} = 7.79$$

FINAL ANSWER

7.79

PAST PAPER QUESTION**Q#: 19****Marks: 3****TOPIC: Algebraic Fractions**

Jun 2022 P2H - Jun 2022 | P2H

19 $a = \frac{14}{3x-7}$ $x = \frac{7}{4y-3}$

Express a in the form $\frac{py+q}{ry+s}$ where p , q , r and s are integers.

Give your answer in its simplest form.

$a = \dots\dots\dots$

(Total for Question 19 is 3 marks)

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PAST PAPER SOLUTION**Q#: 19****Marks: 3****TOPIC: Algebraic Fractions**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Algebraic fractions. The tag is correct.**Solution**

$$a = \frac{14}{3x - 7}, \quad x = \frac{7}{4y - 3}$$

Substitute for x :

$$\begin{aligned} 3x - 7 &= \frac{21}{4y - 3} - 7 \\ &= \frac{21 - 7(4y - 3)}{4y - 3} = \frac{42 - 28y}{4y - 3} \end{aligned}$$

So

$$a = \frac{14}{\frac{42 - 28y}{4y - 3}} = \frac{4y - 3}{3 - 2y}$$

FINAL ANSWER

$$a = \frac{4y - 3}{3 - 2y}$$

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Jun 2022 P2H - Jun 2022 | P2H

20 The diagram shows four identical circles drawn inside a square.

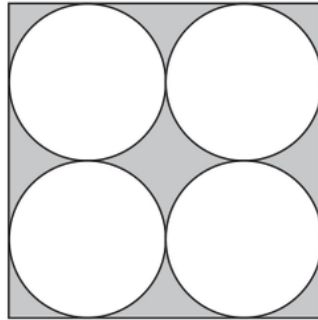


Diagram **NOT**
accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of 40 cm^2

Work out the perimeter of the square.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved to Circles, Arcs & Sectors.**Method**Let the radius of each circle be r .

There are two circles across the width of the square, so the side length of the square is

$$4r$$

Area of the square:

$$(4r)^2 = 16r^2$$

Area of the four circles:

$$4\pi r^2$$

The shaded area is 40 cm^2 , so

$$16r^2 - 4\pi r^2 = 40$$

$$r^2(16 - 4\pi) = 40$$

$$r = \sqrt{\frac{40}{16 - 4\pi}} = 3.4131 \dots$$

The perimeter of the square is

$$4(4r) = 16r$$

$$16r = 54.6101 \dots$$

FINAL ANSWER**54.6 cm**

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Jun 2022 P2H - Jun 2022 | P2H

20 The diagram shows four identical circles drawn inside a square.

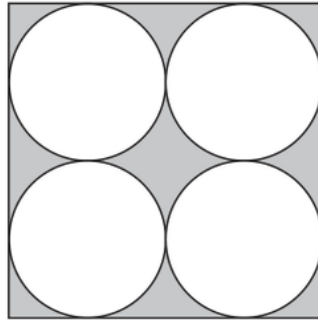


Diagram **NOT**
accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of 40 cm^2

Work out the perimeter of the square.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved to Circles, Arcs & Sectors.**Method**Let the radius of each circle be r .

There are two circles across the width of the square, so the side length of the square is

$$4r$$

Area of the square:

$$(4r)^2 = 16r^2$$

Area of the four circles:

$$4\pi r^2$$

The shaded area is 40 cm^2 , so

$$16r^2 - 4\pi r^2 = 40$$

$$r^2(16 - 4\pi) = 40$$

$$r = \sqrt{\frac{40}{16 - 4\pi}} = 3.4131 \dots$$

The perimeter of the square is

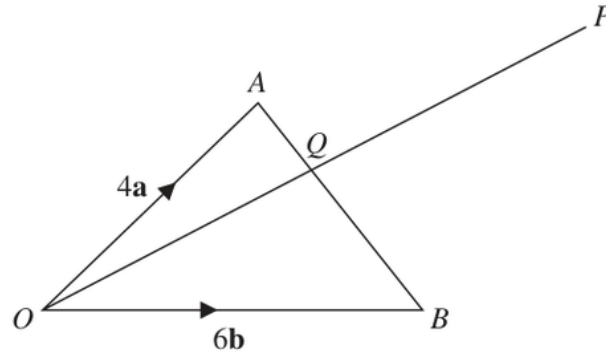
$$4(4r) = 16r$$

$$16r = 54.6101 \dots$$

FINAL ANSWER**54.6 cm**

PAST PAPER QUESTION**Q#: 21****Marks: 5**TOPIC: **Vectors**

Jun 2022 P2H - Jun 2022 | P2H

21Diagram **NOT**
accurately drawn OAB is a triangle. Q is the point on AB such that OQP is a straight line.

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio $AQ:QB$

$$AQ:QB = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 5**TOPIC: **Vectors**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Ratio Toolkit to Vectors.**Method**

$$\overrightarrow{OA} = 4a, \quad \overrightarrow{OB} = 6b, \quad \overrightarrow{AP} = 2a + 8b$$

So

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = 4a + (2a + 8b) = 6a + 8b$$

Since Q lies on OP , let

$$\overrightarrow{OQ} = \lambda(6a + 8b)$$

Since Q lies on AB , let

$$\overrightarrow{OQ} = \overrightarrow{OA} + \mu\overrightarrow{AB}$$

$$\overrightarrow{AB} = 6b - 4a$$

So

$$\overrightarrow{OQ} = 4a + \mu(6b - 4a) = (4 - 4\mu)a + 6\mu b$$

Equating coefficients:

$$6\lambda = 4 - 4\mu, \quad 8\lambda = 6\mu$$

From $8\lambda = 6\mu$, $\mu = \frac{4}{3}\lambda$.

$$6\lambda = 4 - \frac{16}{3}\lambda$$

$$34\lambda = 12, \quad \lambda = \frac{6}{17}$$

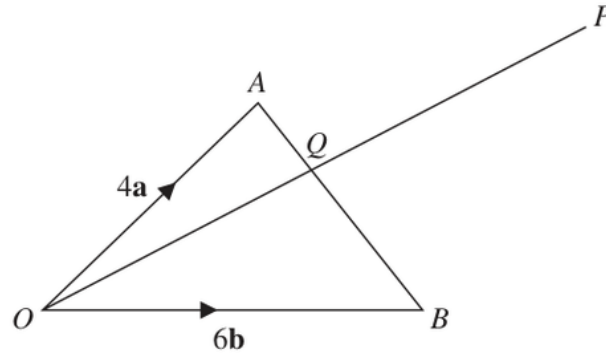
$$\mu = \frac{8}{17}$$

So $AQ : QB = \frac{8}{17} : \frac{9}{17}$.**FINAL ANSWER**

$$AQ : QB = 8 : 9$$

PAST PAPER QUESTION**Q#: 21****Marks: 5**TOPIC: **Vectors**

Jun 2022 P2H - Jun 2022 | P2H

21Diagram **NOT**
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$$AQ:QB = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 5**TOPIC: **Vectors**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Ratio Toolkit to Vectors.**Method**

$$\overrightarrow{OA} = 4a, \quad \overrightarrow{OB} = 6b, \quad \overrightarrow{AP} = 2a + 8b$$

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$$AQ : QB = 8 : 9$$

PAST PAPER QUESTION**Q#: 22****Marks: 7**TOPIC: **Coordinate Geometry**

Jun 2022 P2H - Jun 2022 | P2H

- 22** $ABCD$ is a kite, with diagonals AC and BD , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,

find the coordinates of the points B and D .

(.....,)

(.....,)

(Total for Question 22 is 7 marks)

Dr. Eslam

PAST PAPER SOLUTION**Q#: 22****Marks: 7****TOPIC: Coordinate Geometry**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**The midpoint of BD is $M = (0, 2)$.Since $A = (-3, 4)$, the gradient of AM is

$$\frac{2 - 4}{0 - (-3)} = -\frac{2}{3}$$

For the kite, diagonal AC is perpendicular to BD , so the gradient of BD is

$$\frac{3}{2}$$

Use direction vector $(2, 3)$ along BD .

Let

$$B = (2t, 2 + 3t)$$

and

$$D = (-2t, 2 - 3t)$$

Since $AB = 6.5$,

$$(2t - (-3))^2 + (2 + 3t - 4)^2 = 6.5^2$$

$$(2t + 3)^2 + (3t - 2)^2 = 42.25$$

$$4t^2 + 12t + 9 + 9t^2 - 12t + 4 = 42.25$$

$$13t^2 + 13 = 42.25$$

$$13t^2 = 29.25$$

$$t^2 = 2.25$$

$$t = 1.5$$

The x -coordinate of B is positive, so use $t = 1.5$.

$$B = (3, 6.5)$$

$$D = (-3, -2.5)$$

FINAL ANSWER

$$B = (3, 6.5), D = (-3, -2.5)$$

Dr. Eslam Ahmed

PAST PAPER QUESTION**Q#: 22** **Marks: 7****TOPIC: Coordinate Geometry**

Jun 2022 P2H - Jun 2022 | P2H

22 $ABCD$ is a kite, with diagonals AC and BD , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

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(Total for Question 22 is 7 marks)

Dr. Eslam

PAST PAPER SOLUTION**Q#: 22****Marks: 7**TOPIC: **Coordinate Geometry**

Jun 2022 P2H - Jun 2022 | P2H

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FINAL ANSWER

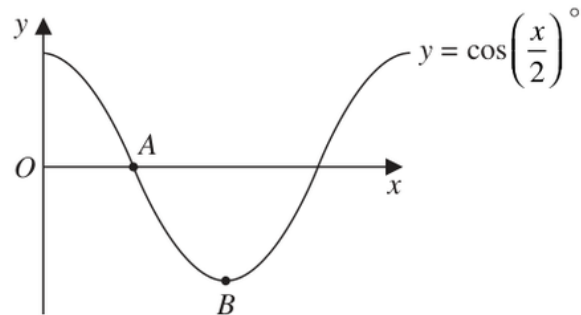
$$B = (3, 6.5), D = (-3, -2.5)$$

Dr. Eslam Ahmed

PAST PAPER QUESTION**Q#: 23****Marks: 2****TOPIC: Graphs of Functions**

Jun 2022 P2H - Jun 2022 | P2H

- 23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



- (i) Find the coordinates of the point A

(.....,)
(1)

- (ii) Find the coordinates of the point B

(.....,)
(1)

(Total for Question 23 is 2 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 2****TOPIC: Graphs of Functions**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved from Coordinate Geometry to Graphs of Functions.**Method**

The graph is

$$y = \cos\left(\frac{x}{2}\right)^\circ$$

Point A is the first positive x -intercept, so $y = 0$.

$$\cos\left(\frac{x}{2}\right) = 0$$

$$\frac{x}{2} = 90^\circ$$

$$x = 180$$

So

$$A = (180, 0)$$

Point B is the minimum point.The minimum value of cosine is -1 , which occurs when

$$\frac{x}{2} = 180^\circ$$

$$x = 360$$

So

$$B = (360, -1)$$

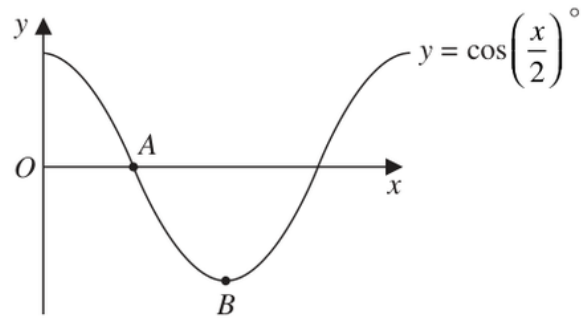
FINAL ANSWER

$$A = (180, 0), B = (360, -1)$$

PAST PAPER QUESTION**Q#: 23****Marks: 2****TOPIC: Graphs of Functions**

Jun 2022 P2H - Jun 2022 | P2H

- 23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



- (i) Find the coordinates of the point A

(.....,)
(1)

- (ii) Find the coordinates of the point B

(.....,)
(1)

(Total for Question 23 is 2 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 2****TOPIC: Graphs of Functions**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved from Coordinate Geometry to Graphs of Functions.**Method**

The graph is

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Point B is the minimum point.The minimum value of cosine is -1 , which occurs when

$$\frac{x}{2} = 180^\circ$$

$$x = 360$$

So

$$B = (360, -1)$$

FINAL ANSWER

$$A = (180, 0), B = (360, -1)$$

PAST PAPER QUESTION**Q#: 24****Marks: 3**TOPIC: **Algebraic Roots & Indices**

Jun 2022 P2H - Jun 2022 | P2H

24

$$\frac{18 \times (\sqrt{27})}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$ **(Total for Question 24 is 3 marks)**

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 24****Marks: 3****TOPIC: Algebraic Roots & Indices**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Algebraic Roots and Indices. The tag is correct.**Method**

$$\frac{18(\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

First,

$$\frac{18}{6} = 3 = 3^1$$

Also,

$$\sqrt{27} = \sqrt{3^3} = 3^{3/2}$$

So

$$(\sqrt{27})^{4n+6} = 3^{\frac{3}{2}(4n+6)} = 3^{6n+9}$$

And

$$9^{2n+8} = (3^2)^{2n+8} = 3^{4n+16}$$

Therefore the whole expression is

$$3^{1+6n+9-(4n+16)} = 3^{2n-6}$$

So

$$x = 2n - 6$$

FINAL ANSWER

$$x = 2n - 6.$$

PAST PAPER QUESTION**Q#: 24****Marks: 3****TOPIC: Algebraic Roots & Indices**

Jun 2022 P2H - Jun 2022 | P2H

24

$$\frac{18 \times (\sqrt{27})}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$ **(Total for Question 24 is 3 marks)**

Dr. Eslam Ahmed

PAST PAPER SOLUTION**Q#: 24****Marks: 3**TOPIC: **Algebraic Roots & Indices**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$\frac{18(\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Use powers of 3:

$$18 \div 6 = 3$$

$$\sqrt{27} = (3^3)^{1/2} = 3^{3/2}$$

$$(\sqrt{27})^{4n+6} = 3^{6n+9}$$

$$9^{2n+8} = (3^2)^{2n+8} = 3^{4n+16}$$

So

$$\frac{3 \cdot 3^{6n+9}}{3^{4n+16}} = 3^{1+6n+9-4n-16} = 3^{2n-6}$$

Therefore,

$$x = 2n - 6$$

FINAL ANSWER

$$x = 2n - 6$$